

C:\Users\paulk\Documents\Projects\IBM-1620\retro-1620\software\IPL-V\IPL-V-Interpreter-Mod-3-4\IPL-V-Punch-Trace-Ack-1-0-20260823.txt										C:\Users\paulk\Documents\Projects\IBM-1620\retro-1620\software\IPL-V\IPL-V-Interpreter-Mod-3-4\IPL-V-Punch-Trace-Ack-1-1-20260823.txt																													
1	Ackermann(1,0) Trace									1	Ackermann(1,1) Trace																												
2										2																													
3	0001	ACKERMANN-S FUNCTION IN IPL-V				9				3	0001	ACKERMANN-S FUNCTION IN IPL-V				9																							
4	0002	1620 IPL-V ASSEMBLY LISTING				4	ADDRESS WORD LC			4	0002	1620 IPL-V ASSEMBLY LISTING				4	ADDRESS WORD LC																						
5	0003	GIVEN M AND N, COMPUTE A(M,N)				1				5	0003	GIVEN M AND N, COMPUTE A(M,N)				1																							
6	0004	A(0,N) = N + 1				1				6	0004	A(0,N) = N + 1				1																							
42	0040	M+1 TO M. EXIT WITH J8=30H0					00J125	J8	19223	00169551	42	0040	M+1 TO M. EXIT WITH J8=30H0					00J125	J8	19223	00169551																		
43	0041					1					43	0041					1																						
44	0042					5	01					44	0042					5	01																				
45	0043	CONSTANT 1					K1	01	1	18647 01000001	45	0043	CONSTANT 1					K1	01	1	18647 01000001																		
46	0044	VALUE OF M					M0	01	1	18587 01000001	46	0044	VALUE OF M					M0	01	1	18587 01000001																		
47	0045	VALUE OF N					N0	01	0	18611 01000000	47	0045	VALUE OF N					N0	01	1	18611 01000001																		
48	0046	START CARD. EXECUTE E0				5	E0					48	0046	START CARD. EXECUTE E0				5	E0																				
49	0047	SYSTEM SUBROUTINE HEADER CARD				5	19	9 9 9				49	0047	SYSTEM SUBROUTINE HEADER CARD				5	19	9 9 9																			
50	0048						W10	9-1	0	18179 0219247	50	0048						W10	9-1	0	18179 0219247																		
51	0049						0 9-1	+0172459	06183	19247 01724590	51	0049						0 9-1	+0172459	06183	19247 01724590																		
52	0050						W20	9-1	0	18299 0219259	52	0050						W20	9-1	0	18299 0219259																		
78	2 E				MONITOR POINT J										78	2 E				MONITOR POINT J																			
79	3 E				0 3 A				18983	A	0 00000							79	3 E				0 3 A				18983	A	0 00000										
80	4 A				1 4 M				19007	M	0 00000							80	4 A				1 4 M				19007	M	0 00000										
81	4 19007				0 0 J117				19019	J117	4 M	01	10000							81	4 19007				0 0 J117				19019	J117	4 M	01	10000						
82	4 19019				- 1 0 N				19031	N	4 M	01	10000							82	4 19019				- 1 0 N				19031	N	4 M	01	10000						
83	4 19031				- 7 0 19055				19043	19055	0 N	01	0							83	4 19031				- 7 0 19055				19043	19055	0 N	01	10000						
																			84	4 19055				- 0 0 J117				19067	J117	0 N	01	10000							
																			85	4 19067				- 7 0 19091				19079	19091	0 N	01	10000							
																			86	4 19091				- 1 0 K1				19115	K1	0 N	01	10000							
																			87	4 19115				- 1 0 N				19139	N	0 K1	01	10000							
																			88	4 19139				- 1 0 N				19151	N	0 N	01	10000							
																			89	4 19151				- 0 0 J111				19163	J111	0 N	01	10000							
																			90	4 19163				- 0 0 A				19127	A	0 N	01	0							
																			91	5 A				- 1 4 M				19007	M	0 N	01	0							
																			92	5 19007				- 0 0 J117				19019	J117	4 M	01	10000							
																			93	5 19019				- 1 0 N				19031	N	0 N	01	0							
																			94	5 19031				- 7 0 19055				19043	19055	0 N	01	0							
																			95	5 19055				- 0 0 J117				19067	J117	0 N	01	0							
																			96	5 19067				+ 7 0 19091				19079	19091	0 N	01	0							
																			97	5 19079				+ 1 0 N				19103	N	0 N	01	0							
																			98	5 19103				+ 0 0 J125				19127	J125	0 N	01	0							
																			99	5 19127				+ 5 0 K1				19175	K1	0 N	01	1							
																			100	5 19175				+ 1 0 M				19187	M	0 K1	01	10000							
																			101	5 19187				+ 1 0 M				19199	M	0 M	01	10000							
																			102	5 19199				+ 0 0 J111				19211	J111	0 M	01	10000							
																			103	5 19211				+ 0 0 A				19223	A	0 M	01	0							
																			104	6 A				+ 1 4 M				19007	M	0 M	01	0							
																			105	6 19007				+ 0 0 J117				19019	J117	4 M	01	0							

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88	4	19127	+ 5 0 K1	19175	K1	0 N	01	1	106	6	19019	+ 1 0 N	19031	N	0 M	01	0
89	4	19175	+ 1 0 M	19187	M	0 K1	01	10000	107	6	19031	+ 7 0 19055	19043	19055	0 N	01	1
90	4	19187	+ 1 0 M	19199	M	0 M	01	10000	108	6	19043	+ 0 0 J125	J8	J125	0 N	01	1
91	4	19199	+ 0 0 J111	19211	J111	0 M	01	10000	109	6	J8	+ 0 5 05578	0	05578	0 N	01	2
92	4	19211	+ 0 0 A	19223	A	0 M	01	0	110	5	19223	+ 0 0 J125	J8	J125	0 M	01	0
93	5	A	+ 1 4 M	19007	M	0 M	01	0	111	5	J8	+ 0 5 05578	0	05578	0 M	01	1
									112	4	19127	+ 5 0 K1	19175	K1	0 N	01	2
									113	4	19175	+ 1 0 M	19187	M	0 K1	01	10000
									114	4	19187	+ 1 0 M	19199	M	0 M	01	1
									115	4	19199	+ 0 0 J111	19211	J111	0 M	01	1
									116	4	19211	+ 0 0 A	19223	A	0 M	01	-9999
									117	5	A	+ 1 4 M	19007	M	0 M	01	-9999
									118	5	19007	+ 0 0 J117	19019	J117	4 M	01	-9999
									119	5	19019	- 1 0 N	19031	N	0 M	01	-9999
									120	5	19031	- 7 0 19055	19043	19055	0 N	01	2
									121	5	19055	- 0 0 J117	19067	J117	0 N	01	2
									122	5	19067	- 7 0 19091	19079	19091	0 M	01	-9999
									123	5	19091	- 1 0 K1	19115	K1	0 M	01	-9999
									124	5	19115	- 1 0 N	19139	N	0 K1	01	10000
									125	5	19139	- 1 0 N	19151	N	0 N	01	2
									126	5	19151	- 0 0 J111	19163	J111	0 N	01	2
									127	5	19163	- 0 0 A	19127	A	0 N	01	-9998
									128	6	A	- 1 4 M	19007	M	0 N	01	-9998
									129	6	19007	- 0 0 J117	19019	J117	4 M	01	-9999
									130	6	19019	- 1 0 N	19031	N	0 N	01	-9998
									131	6	19031	- 7 0 19055	19043	19055	0 N	01	-9998
									132	6	19055	- 0 0 J117	19067	J117	0 N	01	-9998
									133	6	19067	- 7 0 19091	19079	19091	0 N	01	-9998
									134	6	19091	- 1 0 K1	19115	K1	0 N	01	-9998
									135	6	19115	- 1 0 N	19139	N	0 K1	01	10000
									136	6	19139	- 1 0 N	19151	N	0 N	01	-9998
									137	6	19151	- 0 0 J111	19163	J111	0 N	01	-9998
									138	6	19163	- 0 0 A	19127	A	0 N	01	-19998
									139	7	A	- 1 4 M	19007	M	0 N	01	-19998
									140	7	19007	- 0 0 J117	19019	J117	4 M	01	-9999
									141	7	19019	- 1 0 N	19031	N	0 N	01	-19998
									142	7	19031	- 7 0 19055	19043	19055	0 N	01	-19998
									143	7	19055	- 0 0 J117	19067	J117	0 N	01	-19998
									144	7	19067	- 7 0 19091	19079	19091	0 N	01	-19998
									145	7	19091	- 1 0 K1	19115	K1	0 N	01	-19998
									146	7	19115	- 1 0 N	19139	N	0 K1	01	10000
									147	7	19139	- 1 0 N	19151	N	0 N	01	-19998
									148	7	19151	- 0 0 J111	19163	J111	0 N	01	-19998
									149	7	19163	- 0 0 A	19127	A	0 N	01	-29998
									150	8	A	- 1 4 M	19007	M	0 N	01	-29998
94	5	19007	+ 0 0 J117	19019	J117	4 M	01	0	151	8	19007	- 0 0 J117	19019	J117	4 M	01	-9999
95	5	19019	+ 1 0 N	19031	N	0 M	01	0	152	8	19019	- 1 0 N	19031	N	0 N	01	-29998
96	5	19031	+ 7 0 19055	19043	19055	0 N	01	1	153	8	19031	- 7 0 19055	19043	19055	0 N	01	-29998
97	5	19043	+ 0 0 J125	J8	J125	0 N	01	1	154	8	19055	- 0 0 J117	19067	J117	0 N	01	-29998
98	5	J8	+ 0 5 05578	0	05578	0 N	01	2	155	8	19067	- 7 0 19091	19079	19091	0 N	01	-29998
99	4	19223	+ 0 0 J125	J8	J125	0 M	01	0	156	8	19091	- 1 0 K1	19115	K1	0 N	01	-29998
100	4	J8	+ 0 5 05578	0	05578	0 M	01	1	157	8	19115	- 1 0 N	19139	N	0 K1	01	10000

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101	3	18983		+	1	0	N	18995	N	0	M	01		1		158		8	19139	-	1	0	N	19151	N	0	N	01		-29998
102	3	18995		+	0	0	J152	0	J152	0	N	01		2		159		8	19151	-	0	0	J111	19163	J111	0	N	01		-29998
103			N				01							2		160		8	19163	-	0	0	A	19127	A	0	N	01		-39998
104	3	18995					MONITOR	END	J							161		9	A	-	1	4	M	19007	M	0	N	01		-39998
105	2	18995					THE	END								162		9	19007	-	0	0	J117	19019	J117	4	M	01		-9999
																163		9	19019	-	1	0	N	19031	N	0	N	01		-39998
																164		9	19031	-	7	0	19055	19043	19055	0	N	01		-39998
																165		9	19055	-	0	0	J117	19067	J117	0	N	01		-39998
																166		9	19067	-	7	0	19091	19079	19091	0	N	01		-39998
																167		9	19091	-	1	0	K1	19115	K1	0	N	01		-39998
																168		9	19115	-	1	0	N	19139	N	0	K1	01		10000
																169		9	19139	-	1	0	N	19151	N	0	N	01		-39998
																170		9	19151	-	0	0	J111	19163	J111	0	N	01		-39998
																171		9	19163	-	0	0	A	19127	A	0	N	01		-49998
																172	10	A		-	1	4	M	19007	M	0	N	01		-49998
																173	10	19007		-	0	0	J117	19019	J117	4	M	01		-9999
																174	10	19019		-	1	0	N	19031	N	0	N	01		-49998
																175	10	19031		-	7	0	19055	19043	19055	0	N	01		-49998
																176	10	19055		-	0	0	J117	19067	J117	0	N	01		-49998
																177	10	19067		-	7	0	19091	19079	19091	0	N	01		-49998
																178	10	19091		-	1	0	K1	19115	K1	0	N	01		-49998
																179	10	19115		-	1	0	N	19139	N	0	K1	01		10000
																180	10	19139		-	1	0	N	19151	N	0	N	01		-49998
																181	10	19151		-	0	0	J111	19163	J111	0	N	01		-49998
																182	10	19163		-	0	0	A	19127	A	0	N	01		-59998
																183	11	A		-	1	4	M	19007	M	0	N	01		-59998
																184	11	19007		-	0	0	J117	19019	J117	4	M	01		-9999
																185	11	19019		-	1	0	N	19031	N	0	N	01		-59998
																186	11	19031		-	7	0	19055	19043	19055	0	N	01		-59998
																187	11	19055		-	0	0	J117	19067	J117	0	N	01		-59998
																188	11	19067		-	7	0	19091	19079	19091	0	N	01		-59998
																189	11	19091		-	1	0	K1	19115	K1	0	N	01		-59998
																190	11	19115		-	1	0	N	19139	N	0	K1	01		10000
																191	11	19139		-	1	0	N	19151	N	0	N	01		-59998
																192	11	19151		-	0	0	J111	19163	J111	0	N	01		-59998
																193	11	19163		-	0	0	A	19127	A	0	N	01		-69998
																194	12	A		-	1	4	M	19007	M	0	N	01		-69998
																195	12	19007		-	0	0	J117	19019	J117	4	M	01		-9999
																196	12	19019		-	1	0	N	19031	N	0	N	01		-69998
																197	12	19031		-	7	0	19055	19043	19055	0	N	01		-69998
																198	12	19055		-	0	0	J117	19067	J117	0	N	01		-69998
																199	12	19067		-	7	0	19091	19079	19091	0	N	01		-69998
																200	12	19091		-	1	0	K1	19115	K1	0	N	01		-69998
																201	12	19115		-	1	0	N	19139	N	0	K1	01		10000
																202	12	19139		-	1	0	N	19151	N	0	N	01		-69998
																203	12	19151		-	0	0	J111	19163	J111	0	N	01		-69998
																204	12	19163		-	0	0	A	19127	A	0	N	01		-79998
																205	13	A		-	1	4	M	19007	M	0	N	01		-79998
																206	13	19007		-	0	0	J117	19019	J117	4	M	01		-9999
																207	13	19019		-	1	0	N	19031	N	0	N	01		-79998
																208	13	19031		-	7	0	19055	19043	19055	0	N	01		-79998
																209	13	19055		-	0	0	J117	19067	J117	0	N	01		-79998